AeroCast: A Leakage-Checked, Calibrated Flight Delay Prediction Framework with Schedule-Time and Two-Hour-Ahead Modes

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Abstract—Published flight-delay results are hard to compare because splits, horizons, delay definitions and leakage controls differ. We study departure-delay prediction on 1,212,998 non-cancelled US flights from the Bureau of Transportation Statistics (BTS, June–July 2026) under one chronological train/validation/test protocol, giving every model identical features. Two operating modes are defined by the information horizon: a schedule-time mode, and a mode issued two hours before departure that uses delays already observed. Seven learners and four rule-based baselines are compared. On the two-hour-ahead features, gradient-boosted trees, a random forest and neural models are practically equivalent (within about 0.003 AUC; LightGBM 0.743, 95% day-block CI [0.737, 0.749]), while logistic regression is lower (0.722). Leakage-checked features, not the learner, drive the gain: AUC rises +0.026 (CI [0.021, 0.031]) over a 15-feature XGBoost baseline (0.717), and +0.006 for the schedule-time mode; gains hold in four rolling-origin windows. The final LightGBM model is already well calibrated before post-processing (ECE 0.014, Brier 0.173); isotonic calibration fitted on one validation week does not improve it (ECE 0.021, Brier 0.173), whereas it greatly helps a class-weighted baseline (ECE 0.194 → 0.044). At a validation-chosen threshold the model attains precision 0.454, recall 0.700 and F1 0.551. A random split overstates AUC by about 0.024. Validation covers a single summer, delay duration remains weakly predictable and weather was not evaluated. Code, models and checksums are released.

***Index Terms*—flight delay prediction, calibration, data leakage, benchmarking, gradient boosting, explainable machine learning.**

I. Introduction

Flight delays are costly and propagate through airline and airport networks, and reviews of the propagation literature cover machine-learning prediction as a main approach [1]. Reported performance ranges widely (for example 85–92% departure-delay accuracy in [5] versus a 19% error at a 60-minute threshold in [3]) because the studies differ in data, horizon, target and validation. This paper asks four questions:

RQ1. Under a single fair protocol, how do strong baselines compare, and does the learner or the feature set matter more? **RQ2.** How much does information available two hours before departure add to schedule-time prediction? **RQ3.** Are probabilities calibrated, and how wide are honest confidence intervals? **RQ4.** How large is the optimism of random splitting?

Contribution. We do not propose a new learning algorithm; the models, SHAP explanations and upstream-delay features follow prior work [3]–[6], [12]–[14]. Our contribution is a leakage-checked, two-horizon prediction framework and benchmark: (1) a same-split, same-feature benchmark of seven learners and four rule baselines (including a neural-embedding-plus-trees hybrid in the style of [5]) on 1.2 million public BTS flights, showing that leading learners are equivalent and features drive the gain; (2) two explicit information-horizon modes with documented feature-availability timestamps and brute-force-verified leakage controls (Section IV-B); (3) calibrated probabilities, validation-chosen thresholds, risk bands, and day-block bootstrap intervals and paired differences, which are rarely reported in this literature; (4) rolling-origin evaluation and a measured random-split inflation; and (5) an openly reported set of negative results and a reproducibility package.

II. Related Work

Delay propagation and networks. Li and Yao [1] review propagation within airlines (econometric models, Bayesian networks, propagation trees) and between airports, noting that machine learning is the most common approach for predicting delays within airlines and that deep and spatio-temporal models improve inter-airport prediction accuracy. Rebollo and Balakrishnan [3] introduced Random Forest models using local and network-wide delay states to predict delays on origin–destination links 2–24 hours ahead; on 2007–2008 ASPM data, at a 2-hour horizon the average test error over the 100 most delayed links was 19.1% for a 60-minute threshold and the median regression error was about 21 minutes. Our two-hour-ahead mode follows this idea with simpler aggregate delay-state features.

Factor-based and deep models. Wang et al. [2] use an LSTM with attention to separate direct from time-related indirect factors, reporting a mean absolute error of about 8.15 minutes on Beijing airport operational data. Yu et al. [4] combine a deep belief network with support vector regression on Beijing airport data. Jha et al. [5] feed activations of a fully connected network to tree ensembles, using 54,000 BTS records (2,000 randomly sampled per month, January 2018–March 2020) split into 40,500 training and 13,500 test records, and report departure-delay accuracies from 85.6% (random forest) to 91.6% (XGBoost). Zhou [6] proposes a two-stage framework (CatBoost estimates a delay-absorption score, XGBoost predicts late departures) on summer-2023 US domestic flights with NOAA weather, reporting ROC-AUC 0.898 against 0.865 for the baseline; Zhang et al. [7] use a stacked LSTM for one-hour-ahead arrival-delay regression at Atlanta.

Limits of schedule information and airport studies. Murad and Ruocco [8] show on about 1.7 million European flights

that, with scheduled information only, R^2 stays at or below 0.44 for the targets studied (peaking at 0.44 for turnaround time and 0.36 for departure delay), a realistic ceiling for pre-tactical prediction. Lemay and Bastin [9] compare logistic regression, random forests and gradient boosting at one major international airport (652,781 flights, 2019–2025, 36-hour horizon) and highlight historical delay rates of flight numbers and airlines.

Gap. The cited studies differ in horizon, target threshold, data period and validation (random sampling or random record splits in [5], [8], [9]), seldom report calibration or interval estimates, and rarely compare strong learners on identical features. It is therefore unclear whether reported gains come from the learner, the features, or the protocol. Table I summarises study designs; scores are listed as reported and are not directly comparable.

TABLE I. Prior work as reported (design, not raw accuracy)

Study	Data	Horizon	Delay definition	Model	Validation	Metric reported
Rebollo & Balakrishnan [3]	ASPM, 2007–08; 100 most delayed OD pairs	2–24 h	60-min threshold (classification); minutes (regression)	Random Forest	Ten random train/test samples (3,000/1,000 points)	2 h: 19.1% error; median error about 21 min
Jha et al. [5]	BTS, Jan 2018–Mar 2020; 54k sampled records	Not stated in text checked	Not stated in text checked	FCNN + RF / XGBoost	40,500 / 13,500 split of a randomly sampled subset	Departure accuracy 85.6–91.6%
Murad & Ruocco [8]	1.74M EU flights, four months of 2019	Hours to days (scheduled info only)	Minutes (no threshold)	Five regressors (DT, RF, GB, XGB, CatBoost)	Random 80/20 split	$R^2 \leq 0.44$ (departure delay peak 0.36)
Yu et al. [4]	Beijing airport	—	—	DBN-SVR	—	—
Wang et al. [2]	Beijing airport operations	Not stated in text checked	Minutes (regression)	LSTM with attention	Not stated in text checked	MAE about 8.15 min
Zhang et al. [7]	ATL arrivals, first day of each month Jul 2016–Feb 2017	1 h	Minutes (regression)	Stacked LSTM	Train/test split not reported	MSE only (e.g., 25.6 min² best)
Zhou [6]	US domestic, summer 2023, NOAA weather	Not stated in text checked	Late if > 15 min	CatBoost + XGBoost (two-stage)	Not stated in text checked	ROC-AUC 0.865 → 0.898
Lemay & Bastin [9]	One major intl. airport, 2019–2025 (excl. Apr 2020–Jul 2021), 652,781 flights	Up to 36 h	On time if delay ≤ 15 min	LR, RF, gradient boosting	Random 80/10/10 split	Departure weighted F1 0.71; MAE 20.11 min
This work	BTS Jun–Jul 2026, 1.2M flights	Schedule-time; 2 h	DEP_DELAY ≥ 15 min	7 learners, 4 rules	Date-based split, day-block CIs, rolling windows	AUC 0.723 / 0.743; ECE 0.014 (raw)

III. Data

We use two extracts of the BTS On-Time Reporting database [10] downloaded on 21 September 2026: June 2026 (607,577 records) and July 2026 (631,970 records), 1,239,547 in total, with SHA-256 checksums released with the code. Cancelled flights and records without a departure delay are removed (26,549), leaving 1,212,998 flights, 12 carriers, 352 airports and 6,142 routes. A flight is delayed if DEP_DELAY ≥ 15 min (28.0%). Delay is heavy-tailed (median −1 min, mean 19.5, 95th percentile 110, maximum 3,392).

Weather (Open-Meteo [11]) could not be downloaded in the experimental environment; the weather inputs were constants and are excluded, so weather impact is untested (Section IX).

TABLE II. Date-based split (no day straddles a boundary)

Split	Flights	Dates	Delayed
Train	893,554	06-01 – 07-15	27.1%
Validation	117,983	07-16 – 07-21	33.5%
Test	201,461	07-22 – 07-31	28.7%

The validation delay rate (33.5%) differs from the test rate (28.7%); this matters for calibration (Section VI-B). Extending the data to other seasons is left for future work (Section IX).

IV. Methodology

A. Feature sets and information horizons

The v1 baseline uses 15 features (scheduled hour, day of week, month, weekend flag, distance, six historical delay statistics, four constant weather inputs). The schedule-time set A (21 features) drops month and weather and adds the minute of day, scheduled departures at the origin in that hour, and smoothed delay rates over the previous 1, 7, 14 and 21 days for origin, airline, route, airline-at-origin and origin-and-hour, plus arrival-delay rates for the destination and for flights into the origin (smoothing prior 0.25). The two-hour-ahead set C (37 features) adds 16 features summarising flights that had actually departed between six and two hours before the scheduled time: count, mean delay and share delayed ≥ 15 min for the origin, the airline, the airline at the origin, a shorter four-to-two-hour origin window, and all US flights.

B. Leakage controls and audit

Availability rule. For a flight with scheduled departure time t_s, every feature must be computable from records available at prediction time t_p: t_p is the start of the

schedule day for the schedule-time mode, and $t_p = t_s - 120$ min for the two-hour-ahead mode. Table III lists the availability of each feature group.

TABLE III. FEATURE AVAILABILITY TIMESTAMPS

Feature group	Information used	Available at	Rule enforced
Calendar / schedule (hour, minute, day of week, distance, departures in hour)	Published schedule	Any time before departure	None needed
Daily rolling delay rates (1/7/14/21 days)	Outcomes of dates before date D	Start of day D	Strictly earlier dates only
Tactical windows (count, mean delay, share delayed)	Flights already departed	$t_s - 120$ min	Only flights with actual departure $\leq t_s - 120$ min
Target (DEP_DELAY ≥ 15)	Outcome of the flight itself	Never an input	Excluded, including from all aggregates

Verification performed. Daily statistics and tactical windows were verified by brute-force recomputation on random rows (exact match), and a vectorised feature builder reproduces the original implementation exactly. These checks are not yet packaged as automated unit tests in the release (Section IX).

C. Learners and baselines

All learners use the same features, are trained without class weights, and are tuned only on the validation set: LightGBM [12] (63 leaves, learning rate 0.04, early stopping), XGBoost [13], histogram gradient boosting, Random Forest [3] (60 trees, depth 12, 20% bootstrap samples, reduced for compute), logistic regression, an MLP (64-32 units), and an FCNN+LightGBM hybrid that feeds the MLP's second-layer activations to LightGBM, mirroring the design of [5]. Rule baselines are the training hour-of-day delay rate, the 21-day origin-hour rate, and the recent origin and airline delay shares (two-hour-ahead mode), plus a constant prior. The v1 baseline is the original class-weighted XGBoost.

D. Calibration, thresholds and uncertainty

Scores are optionally calibrated by isotonic regression fitted on the validation set; the decision threshold maximises validation F1; precision, recall and F1 are computed on the test set, which is not used for any fitting or selection. Calibration is summarised by the Brier score, expected calibration error (ECE, ten equal-frequency bins), and the slope and intercept of a logistic recalibration. Confidence intervals resample the ten test days with replacement (100 resamples) because flights within a day are correlated; paired differences use the same resamples. Rolling-origin evaluation retrains on all earlier days for four test windows. Table IV lists the methodological issues found and how they were resolved.

TABLE IV. METHODOLOGICAL ISSUES FOUND AND HOW THEY WERE RESOLVED

Issue	Resolution
Row-index split straddled days	Date-based split
Flight-level bootstrap too narrow	Day-block bootstrap
Class-weighted, over-confident scores	Unweighted training + calibration check
Features designed after viewing test	Four rolling-origin windows
Constant weather inputs	Excluded; flagged untested
Month acts as time proxy	Dropped
Mislabeled SHAP file, heuristic fallback	SHAP recomputed; fallback flagged
Dataset provenance, hard-coded paths	Checksums, relative paths, pinned versions
Compute-limited RF/MLP settings	Disclosed

V. SYSTEM ARCHITECTURE

The FastAPI backend serves predictions, analytics and metadata; the Next.js frontend provides the predictor and six dashboards; both run under Docker Compose. The schedule-time mode needs only schedule and history tables. The two-hour-ahead mode needs a live feed of departures, which the current system lacks.

Implementation status. The released predictor.py still uses the v1 feature set and a non-SHAP fallback. It must be updated to load the final trained models and the identical feature builder before serving them; until then, the results describe the experimental pipeline, not the deployed predictor.

VI. EXPERIMENTS

All results use the date-based test set (201,461 flights, 28.7% delayed) unless stated.

A. Strong baselines on identical features

TABLE V. TWO-HOUR-AHEAD FEATURES: DISCRIMINATION (95% DAY-BLOCK CI)

Model	ROC-AUC	PR-AUC
LightGBM (proposed)	0.743 [0.737, 0.749]	0.539 [0.506, 0.570]
XGBoost	0.743 [0.737, 0.749]	0.538 [0.506, 0.569]
MLP (64-32)	0.742 [0.734, 0.748]	0.538 [0.502, 0.570]
Hist. grad. boosting	0.741 [0.734, 0.746]	0.535 [0.503, 0.565]
FCNN+LightGBM hybrid	0.741 [0.733, 0.747]	0.537 [0.499, 0.570]
Random Forest	0.740 [0.733, 0.746]	0.531 [0.499, 0.563]
Logistic regression	0.722 [0.717, 0.729]	0.500 [0.470, 0.534]
Rule: origin-hour rate	0.694 [0.686, 0.703]	0.441 [0.406, 0.478]
Rule: hour-of-day rate	0.677 [0.667, 0.687]	0.408 [0.375, 0.440]
Rule: recent airline delay	0.603 [0.598, 0.609]	0.322 [0.299, 0.340]
Rule: recent origin delay	0.602 [0.593, 0.609]	0.340 [0.315, 0.364]
Constant prior	0.500 [0.499, 0.502]	0.287 [0.267, 0.304]

The leading learners cannot be separated in practice: LightGBM exceeds XGBoost by 0.0003 (CI [−0.000, 0.001]) and exceeds the random forest, histogram boosting, MLP and hybrid by at most 0.0033 AUC (some paired intervals exclude zero, but the differences are operationally negligible). Logistic regression is 0.020 lower (CI [0.017, 0.025]), and the best rule (0.694) is far below. On schedule-time features (set A) all learners lie in 0.713–0.724 (LightGBM 0.723 [0.718, 0.728]; Random Forest 0.724; logistic regression 0.713).

TABLE VI. PRECISION, RECALL, F1 AT THE VALIDATION-CHOSEN THRESHOLD

Model (features)	Prec.	Rec.	F1 [95% CI]
v1 XGBoost (15)	0.420	0.722	0.531 –
LightGBM A (21)	0.428	0.712	0.534 [0.512, 0.553]
LightGBM C (37)	0.454	0.700	0.551 [0.528, 0.571]
XGBoost C	0.466	0.669	0.550 [0.526, 0.570]

Model (features)	Prec.	Rec.	F1 [95% CI]
Random Forest C	0.451	0.696	0.548 [0.524, 0.567]
Logistic reg. C	0.433	0.697	0.534 [0.512, 0.553]

B. Calibration

TABLE VII. CALIBRATION (TEST SET)

Model	Brier raw	Brier cal.	ECE raw	ECE cal.
v1 XGBoost (weighted)	0.221	0.183	0.194	0.044
LightGBM A	0.181	0.181	0.027	0.034
LightGBM C	0.173	0.173	0.014	0.021
Logistic reg. C	0.179	0.181	0.023	0.038
Constant prior	0.205	0.207	–	–

Calibration is clearly necessary for the class-weighted v1 model (raw ECE 0.194 → 0.044; Brier 0.221 → 0.183), which is badly over-confident. For the unweighted LightGBM models it is not beneficial on the test set: the raw LightGBM C model is already near-calibrated (raw slope 0.96, intercept −0.11, ECE 0.014), and isotonic regression leaves the Brier score unchanged (0.173) while ECE rises slightly (0.014 → 0.021); the same pattern appears for LightGBM A and logistic regression. We attribute this to fitting the isotonic map on a single validation week whose delay rate (33.5%) differs from the test rate (28.7%) (Table II); this explanation is a hypothesis, not tested. We therefore report both raw and calibrated scores, and treat isotonic calibration as a safeguard against over-confidence rather than a source of improvement. Alternative calibrators (for example Platt scaling) and a larger calibration window were not evaluated.

Calibrated risk bands on the final model are monotone in observed delay rate: 17.3% delayed in the low band (128,615 flights), 43.8% in the moderate band (56,406) and 66.1% in the high band (16,440). The band thresholds were not validated on later unseen data.

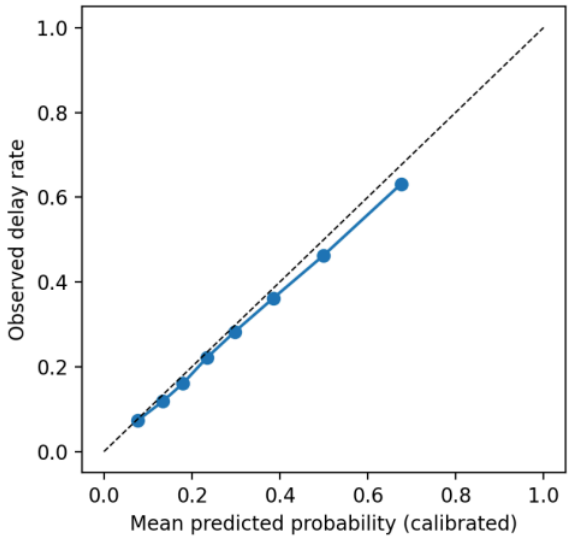


Fig. 1. Calibration of the final two-hour-ahead model (calibrated scores).

C. Paired comparisons and rolling-origin windows

TABLE VIII. PAIRED AUC DIFFERENCES (95% DAY-BLOCK CI)

Comparison	Δ AUC [CI]
Two-hour-ahead C − v1	+0.026 [0.021, 0.031]
Schedule-time A − v1	+0.006 [0.000, 0.012]
C − A	+0.020 [0.016, 0.024]
LightGBM − XGBoost (C)	+0.000 [−0.000, 0.001]
LightGBM − Random Forest (C)	+0.003 [0.002, 0.004]
LightGBM − MLP (C)	+0.002 [0.000, 0.003]
LightGBM − logistic reg. (C)	+0.020 [0.017, 0.025]

TABLE IX. ROLLING-ORIGIN TEST AUC (RETRAINED FOR EACH WINDOW)

Test window	v1 XGB	Sched.-time	2 h-ahead
07-01 – 07-07	0.715	0.721	0.741
07-08 – 07-14	0.722	0.730	0.752
07-15 – 07-21	0.735	0.740	0.759
07-22 – 07-31	0.717	0.723	0.743

The two-hour-ahead model beats the v1 baseline in every window by 0.024–0.030 AUC and the schedule-time model by 0.005–0.008. The features were designed after inspecting the last window only, so the first three windows are less affected by selection.

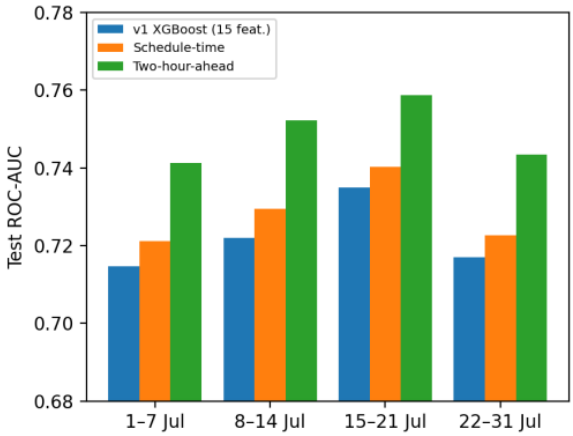


Fig. 2. Rolling-origin test AUC.

D. Delay duration

TABLE X. REGRESSION ON THE TEST SET (MINUTES)

Model / baseline	Eval. on	MAE	RMSE	R^2
LightGBM, all flights	all	28.09	65.10	0.051
Hurdle (p × mean if late)	all	27.29	65.08	0.051
Predict training mean	all	29.89	66.82	−0.000
Predict zero	all	22.67	70.55	−0.115
LightGBM delayed-trained	delayed	53.55	106.41	0.032
LightGBM conditional median	delayed	47.19	110.53	−0.044
Predict training median	delayed	48.50	112.53	−0.082

Duration remains weak: the best R^2 is 0.051, and every mean-based model has worse MAE than predicting zero (22.67 min). Among delayed flights (mean 75.0, median 44 min) a conditional-median model lowers MAE from 48.5 to 47.2 minutes. We therefore keep delay classification as the primary task; the hurdle model in Table X is a first step

toward a two-stage (classify, then estimate duration) design, which we leave for future work. Murad and Ruocco [8] report R^2 of 0.10–0.30 (peak 0.36) for departure delay with scheduled information on European data and a random split; this is higher than our 0.051, but the data, target and split differ, so the gap should not be read as a deficiency of either approach.

VII. Ablations and Attribution

Removing the three national-window features changes AUC by only 0.0007 (CI [−0.000, 0.001]), so the gain does not rely on that time proxy. In the earlier v1 random-forest experiments, removing rolling history lowered AUC from 0.714 to 0.676; scheduled hour alone gave 0.671; and a stratified random split gave 0.736, 0.738 and 0.738 against 0.713 chronologically, an inflation of about 0.024 (mechanism not tested).

Attribution. Mean |SHAP| (log-odds) for the final model is largest for the origin-and-hour 21-day delay rate (0.367), the airline's recent delay share (0.204) and minute of day (0.158); by group, original features 0.350, new schedule-time 0.932, tactical 0.688 (Fig. 3). SHAP values describe the model's reliance on features, not causal effects. A full leave-one-group-out ablation of each feature group is left for future work.

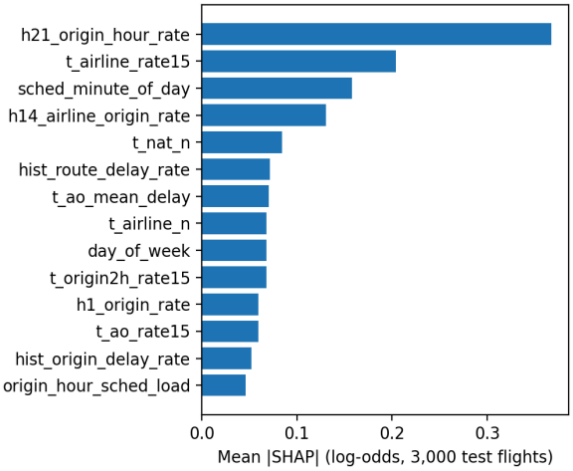


Fig. 3. Mean absolute SHAP value, final model.

VIII. Comparison with Prior Work

What cannot be compared. Reported figures in Table I are not comparable with ours: [3] uses 2007–2008 ASPM data, a 60-minute threshold and the 100 most delayed links; [5] reports accuracy on a 75/25 split of a randomly sampled 54,000-record subset; [8] studies European flights with a different target and a random 80/20 split; [9] uses one airport, a 36-hour horizon and a random 80/10/10 split; and [6] reports ROC-AUC 0.898 on summer-2023 data that include weather and congestion features. Random splits inflate AUC by about 0.024 in our data, so accuracies near 90% under such protocols should not be read as achievable in operation.

What can be compared. The architecture of [5] reproduced on our features reaches AUC 0.741, within the interval of plain LightGBM (0.743); a Random Forest in the spirit of [3] reaches 0.740. Our two-hour-ahead results (AUC 0.743) therefore indicate how much the aggregate delay state helps under a leakage-checked protocol, but a direct comparison with [3] would need their data and threshold. Models using aircraft rotations, networks or deep spatio-temporal structure [1], [2], [6], [7] are outside our feature set, and we do not claim to outperform them.

IX. Discussion and Limitations

Gains come from features known near departure, not from the learner, and are moderate in absolute terms (AUC 0.72 and 0.74). (1) Only 61 days in one summer and a ten-day test window; seasonal behaviour is untested. (2) Features were designed after inspecting the last window, though earlier windows agree. (3) Weather, congestion and aircraft-rotation information are untested; we make no claim about their contribution. (4) The two-hour-ahead mode needs a live departures feed, and the released predictor does not yet match the final models (Section V). (5) Aggregate delays are not aircraft rotations (no tail numbers). (6) Random Forest and MLP settings were reduced for compute, and learners were not exhaustively tuned. (7) Isotonic calibration from one validation week did not improve the final model and may drift in other periods; alternative calibrators and risk-band thresholds were not validated on later data. (8) US domestic flights only. (9) Bootstrap intervals use 100 resamples, and the leakage checks are not yet automated tests.

X. Conclusion

On a fair, calibrated, interval-reported benchmark, the choice among strong learners matters little (within 0.003 AUC), while leakage-checked features raise AUC from 0.717 to 0.723 (schedule-time) and 0.743 (two hours ahead). The final LightGBM model is well calibrated without post-processing. The contribution is a reproducible, leakage-checked benchmark and system with explicit horizons, calibration analysis and negative results, not a new algorithm.

Reproducibility

The release contains scripts (run_all.sh), pinned versions, data checksums, saved models with calibrator and threshold, and result files; all randomness uses fixed seeds.

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